Sizing from a Conceptual Sketch

1 The design process

Design is iterative: *requirements → design concept → design analysis → sizing and trade studies →* back to the requirements.

- **Conceptual design:** Will it work? What does it look like? Which requirements drive the design? What trade-offs? What should it weigh and cost?
- **Preliminary design:** freeze the configuration, develop the test and analytical base, design major items, estimate actual cost.
- **Detail design:** design the pieces to be built, the tooling and fabrication; test major items; finalise weight and performance.

The simplified sizing method is crude, but contains every essential feature of industrial sizing. **It is not suitable for combat or payload-drop missions.**

2 Takeoff weight build-up

W_0 is the *design takeoff gross weight*: the weight at the start of the design mission (not necessarily the maximum takeoff weight).

$$W_0 = W_{crew} + W_{payload} + W_{fuel} + W_{empty}$$

Crew and payload are **known** from the requirements. Fuel and empty weight are **unknown** because they depend on W_0.

$$W_0 = \frac{W_{crew} + W_{payload}}{1 - \dfrac{W_f}{W_0} - \dfrac{W_e}{W_0}}$$

W_e/W_0 depends on W_0, so the equation is solved by **iteration**.

3 Empty weight fraction

Estimated statistically from historical trends. It lies between about 0.3 and 0.7 and *falls* as W_0 increases.

$$\frac{W_e}{W_0} = A\, W_0^{\,C}\, K_{vs} \qquad (W_0 \text{ in lb})$$

Aircraft class	A	C
Sailplane, unpowered	0.86	−0.05
Sailplane, powered	0.91	−0.05
Homebuilt, metal/wood	1.19	−0.09
Homebuilt, composite	0.99	−0.09
General aviation, single engine	2.36	−0.18
General aviation, twin engine	1.51	−0.10
Agricultural aircraft	0.74	−0.03
Twin turboprop	0.96	−0.05
Flying boat	1.09	−0.05
Jet trainer	1.59	−0.10
Jet fighter	2.34	−0.13
Military cargo/bomber	0.93	−0.07
Jet transport	1.02	−0.06

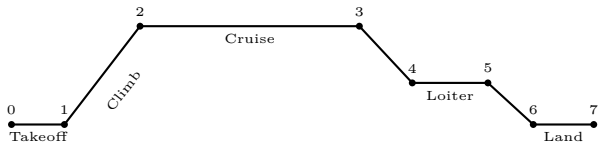
$K_{vs} = 1.04$ for variable sweep, 1.00 for fixed sweep.
Composite structure: take 95% of the metal value, $W_e/W_0 = 0.95\,(A\, W_0^{\,C} K_{vs})$.

4 Fuel fraction

Mission fuel fraction: $1 - W_x/W_0$, where W_x/W_0 is the total mission weight fraction (section 5). Adding 6% for reserve and trapped fuel:

$$\frac{W_f}{W_0} = 1.06\left(1 - \frac{W_x}{W_0}\right)$$

5 Mission profile and segment fractions



Each segment i has a weight fraction W_i/W_{i-1}. Their product is the total mission weight fraction:

$$\frac{W_x}{W_0} = \frac{W_1}{W_0}\,\frac{W_2}{W_1}\,\frac{W_3}{W_2} \cdots \frac{W_x}{W_{x-1}}$$

Segment	W_i/W_{i-1} (historical)
Warm-up and takeoff	0.970
Climb	0.985
Landing	0.995

Descent is treated as part of the cruise range. A loiter of 20 to 30 minutes is usually added for safety.

Cruise (Breguet): $\dfrac{W_i}{W_{i-1}} = \exp\!\left(\dfrac{-R\,C}{V\,(L/D)}\right)$

Loiter (endurance): $\dfrac{W_i}{W_{i-1}} = \exp\!\left(\dfrac{-E\,C}{L/D}\right)$

R range [ft], E endurance [s], V velocity [ft/s], C specific fuel consumption [1/s].

6 Specific fuel consumption, C

Typical jet SFC [1/hr]	Cruise	Loiter
Pure turbojet	0.9	0.8
Low-bypass turbofan	0.8	0.7
High-bypass turbofan	0.5	0.4

Propeller aircraft: $C = C_{bhp}\,\dfrac{V}{550\,\eta_p}$, with V in ft/s.

Typical C_{bhp} / η_p	Cruise	Loiter
Piston-prop, fixed pitch	0.4 / 0.8	0.5 / 0.7
Piston-prop, variable pitch	0.4 / 0.8	0.5 / 0.8
Turboprop	0.5 / 0.8	0.6 / 0.8

7 Estimating L/D

$C_D = C_{D_0} + \dfrac{C_L^2}{\pi\, AR\, e}$: parasite drag scales with **wetted area**, induced drag with **span**. Aspect ratio: $AR = b^2/S$.

$$\text{Wetted aspect ratio} = \frac{b^2}{S_{wet}} = \frac{AR}{S_{wet}/S_{ref}}$$

1. Choose AR; estimate S_{wet}/S_{ref} from the sketch.
2. Calculate the wetted aspect ratio.
3. Read $(L/D)_{max}$ from the chart, using the curve for your aircraft type.
4. Apply the factor for the segment:

	Cruise	Loiter
Jet	$0.866\,(L/D)_{max}$	$(L/D)_{max}$
Prop	$(L/D)_{max}$	$0.866\,(L/D)_{max}$

A much higher AR than comparable aircraft raises L/D but also raises W_e/W_0 above the statistical estimate.

8 Symbols

W_0	takeoff gross weight	W_x	weight at end of mission	R	range	L/D	lift-to-drag ratio
W_e	empty weight	AR	aspect ratio, b^2/S	E	endurance (loiter time)	C	specific fuel consumption
W_f	fuel weight	S_{wet}, S_{ref}	wetted, reference area	V	velocity	K_{vs}, η_p	sweep constant, propeller efficiency